



Entergy Operations, Inc.
River Bend Station
5485 U.S. Highway 61N
St. Francisville, LA 70775
Tel 225-381-4374

Steve Vercelli
Site Vice President

10 CFR 50.73

RBG-48052

October 26, 2020

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2020-03-00, "Weep Holes Discovered in G-Tunnel Wall Result in Inoperable Safety Related Equipment"

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Respectfully,

A handwritten signature in black ink, appearing to read "Steve Vercelli".

SPV/xxx

Enclosure: Licensee Event Report 50-458 / 2020-03-00, "Weep Holes Discovered in G-Tunnel Wall Result in Inoperable Safety Related Equipment"

cc: NRC Regional Administrator - Region IV
NRC Project Manager - River Bend Station
NRC Senior Resident Inspector - River Bend Station
Louisiana Department of Environmental Quality
Public Utility Commission of Texas

Enclosure

RBG-48052

**Licensee Event Report 50-458 / 2020-03-00, "Weep Holes Discovered in G-Tunnel Wall
Result in Inoperable Safety Related Equipment"**



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

River Bend Station

2. Docket Number

05000

458

3. Page

1 OF 2

4. Title

Weep Holes Discovered in G-Tunnel Wall Result in Inoperable Safety Related Equipment

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
8	28	2020	2020	- 03 -	00	10	26	2020	NA	05000 NA
									NA	05000 NA

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☐ 10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	☐ 10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	☐ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER
Licensee Contact

Tim Schenk, Manager - Regulatory Assurance

Phone Number (Include area code)

225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	NA	NA	NA	No	NA	NA	NA	NA	NA

14. Supplemental Report Expected


No



Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

NA

NA

NA

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 5, 2020 at River Bend Station (RBS), weep holes were identified in the wall of the HVAC Chamber located adjacent to the G-Tunnel which could have resulted in the inoperability of safety related equipment. These weep holes were part of the original plant design of the HVAC Chamber. As a result, there was the unknown potential of water infiltration during a Probable Maximum Precipitation (PMP) event. If this infiltration had occurred, there was a cascading potential of an impact to safety related equipment within the G-Tunnel and connecting E and F Tunnels.

This report is made in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event that could have prevented fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
River Bend Station		05000-	458	YEAR 2020	SEQUENTIAL NUMBER 03	REV NO. 00

NARRATIVE**EVENT DESCRIPTION**

On June 10, 2020, weep holes were identified in exterior wall of the HVAC Chamber located adjacent to the G-Tunnel. The HVAC Chamber is a concrete duct which extends from El. 80'-6" where air is admitted through a filter housing (HVY-FLT8) down to El. 66'-3". The bottom of the 30" x 30" ducting to the fan in the G-Tunnel is at 71'-6". Because the original construction had a simple grating at the top (this condition was corrected under MR 93-0086), this vertical shaft could collect precipitation in the bottom. As part of the design documentation four weep holes were to be drilled in the exterior wall at approximately El. 68' to allow standing water to drain to the environment. Only 2 weep holes (each 2 inches in diameter) were confirmed as drilled into the wall.

During a postulated PMP event, water would pond within the excavation area, for what would have been River Bend Unit 2, and would cover these weep holes. During this event, it would have been possible for water to have infiltrated this chamber and, as the water in the excavation area rose, water could have flowed into the ducting for the fan and into the G-Tunnel. A engineering analysis determined that water flow from the weep holes would exceed the capability of the three sump pumps located in the area. The G-Tunnel is interconnected principally with E and F Tunnels and these tunnels contain safety related equipment. As a result, that equipment could have been adversely impacted.

On June 22, 2020 the work was completed to plug the weep holes in accordance with an approved engineering design change.

SAFETY ASSESSMENT

There were no actual consequences to the general safety of the public, nuclear safety, industrial safety and radiological safety as a result of this discovery. A PRA analysis does not exist for external flooding events. Based on the non-mechanistic nature of the PMP, the rainfall amounts exceeding the maximum anticipated yearly rainfall in the Baton Rouge area of Louisiana and the short duration of the event the risk is considered to be low. Additionally, the risk significant equipment (Service Water motor operated valves) in the tunnel complex which would be exposed to this flooding are located several feet above floor level and would likely be in their required positions before seeing any impact from this low probability external flooding scenario.

EVENT CAUSE

The direct cause of this event was due to inadequate design for a ventilation opening.

CORRECTIVE ACTIONS:

RBS took immediate actions to correct the identified conditions. These actions were:

- Completed work order to plug the weep holes
- Revised drawing EC-057P to show the weep holes in the HVAC Chamber for HVY-FN8 as plugged

PREVIOUS SIMILAR EVENTS

None.